Linii de transmisiune

$$\gamma = \sqrt{(R_l + j\omega L_l)(Gl + j\omega C_l)} = \alpha + j\beta$$

$$Z_c = \sqrt{\frac{R_l + j\omega L_l}{G_l + j\omega C_l}}$$

$$\beta = \frac{2\pi}{\lambda}$$

$$v_\varphi = \frac{\omega}{\beta} = \lambda \cdot f$$

$$c = \frac{1}{\sqrt{(\varepsilon_0 \varepsilon_r \mu_0 \mu_r)}} = \frac{c_0}{\sqrt{\varepsilon_r \mu_r}}$$

$$\Gamma = \frac{Z_s - Z_c}{Z_s + Z_c}$$

$$\sigma = \frac{|U_{max}|}{|U_{min}|} = \frac{1 + |\Gamma|}{1 - |\Gamma|} \geq 1$$

$$Z_{in} = Z_c \frac{Z_s + jZ_c tg(\beta \cdot l)}{Z_c + jZ_s tg(\beta \cdot l)}$$

$$\varphi_\Gamma = \arctan\left(\frac{b}{a}\right)$$

$$\alpha = \frac{A}{d}\left[\frac{Np}{m}\right]$$

$$L_l = \frac{\beta^2}{\omega^2 \cdot C_l}\left[\frac{H}{m}\right]$$

$$\Gamma = |\Gamma| \cdot (cosx + jsinx)$$

$$\lambda = \frac{c}{f}$$

$$l = \frac{k\lambda}{4} \Rightarrow Z_{in} = \frac{Z_c^2}{Z_s}, k - impar$$

$$l = \frac{k\lambda}{2} \Rightarrow Z_{in} = Z_s, k - par$$

$$Z_s = R_s \begin{cases} > Z_c \Rightarrow \sigma = \frac{R_s}{Z_c} \\ < Z_c \Rightarrow \sigma = \frac{Z_c}{R_s} \\ = Z_c \Rightarrow \sigma = 1 \end{cases}$$

$$d_{min} = \frac{\pi + \varphi_\Gamma}{4\pi}\lambda + k \cdot \frac{\lambda}{2}; d_{min} \in \left[0, \frac{\lambda}{2}\right)$$

$$\frac{\pi + \varphi_\Gamma}{4\pi}\lambda > 0 \rightarrow k = 0$$

$$d_{max} = \left(\frac{\varphi_\Gamma}{4\pi}\right) \cdot \lambda + k \cdot \frac{\lambda}{2}$$

$$U(z) = U_s\left[\cos(\beta \cdot z) - j\left(\frac{Z_c}{Z_s}\right) \cdot \sin(\beta \cdot z)\right]$$

$$Ps = Pd - Pi = \frac{|U_{d0}|^2}{2Z_c} - \frac{|U_{i0}|^2}{2Z_c} = \frac{|U_{max}|^2}{2\sigma Z_c}$$

$$\frac{Pi}{Pd} = |\Gamma|^2$$

$$P_d = \frac{|U_{d0}|^2}{2Z_c}; P_i = \frac{|U_{i0}|^2}{2Z_c}$$

$$|\Gamma| = \sqrt{a^2 + b^2}, a,b \in \mathbb{R}$$

$$\sqrt{1 + x} = 1 + \frac{x}{2}, x \ll 1$$

Constante

$$\varepsilon_0 = 8.854 \times \frac{10^{-12}F}{m}$$

$$\mu_0 = 4\pi \times \frac{10^{-7}H}{m}$$

$$aer: \varepsilon_r = 1, \mu_r = 1$$

$$c_0 = 3 \times \frac{10^8 m}{s}$$

$$Z_0 = 120\pi\,\Omega = 377\,\Omega$$

$$1\,Np \cong 8.686\,dB$$

Linie fara pierderi $\rightarrow$ $R_l = 0$; $G_l = 0$

Ghid de undă

$$f > f_{c_{m,n}} = \left(\frac{c}{2}\right) \cdot \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$TE_{m,n} = H_{m,n} : m, n \in \mathbb{N};\ m + n \neq 0$$

$$TM_{m,n} = E_{m,n} : m, n \in \mathbb{N}^*$$

$$\lambda_{g_{m,n}} = \frac{\lambda}{\sqrt{1 - \left(\frac{f_{c_{m,n}}}{f}\right)^2}}$$

$$\lambda = \frac{c}{f} = \frac{c_0}{\sqrt{\varepsilon_r} \cdot f} = \frac{\lambda_0}{\sqrt{\varepsilon_r}} \quad (\mu_r = 1)$$

$$\lambda_{c_{m,n}} = \frac{c}{f_{c_{m,n}}}$$

$$v_{f_{m,n}} = \frac{\omega}{\beta_g} = \frac{c}{\sqrt{1 - \left(\frac{f_{c_{m,n}}}{f}\right)^2}}$$

$$v_{g_{m,n}} = c \cdot \sqrt{1 - \left(\frac{f_{c_{m,n}}}{f}\right)^2}$$

$$(Z_u)_{H_{m,n}} = \frac{Z_d}{\sqrt{1 - \left(\frac{f_{c_{m,n}}}{f}\right)^2}} > Z_d$$

$$(Z_u)_{E_{m,n}} = Z_d \cdot \sqrt{1 - \left(\frac{f_{c_{m,n}}}{f}\right)^2} < Z_d$$

$$\gamma_g = \sqrt{k^2 - \omega^2 \varepsilon \mu}$$

$$k = \frac{2\pi}{\lambda_c}$$

$$Zd = \sqrt{\frac{\mu}{\varepsilon}} = \sqrt{\frac{\mu_0}{\varepsilon_0 \varepsilon_r}} = \frac{Z0}{\sqrt{\varepsilon_r}}$$

$$P_{\max,tr} = \frac{|E|^2_{max}}{4\,(Z_u)_{H_{1,0}}}\,ab\frac{1}{\sigma}$$

$$P_{max} = P_{max,tr} \cdot C$$